



Armada Kalamunda Group

Management of Patients Requiring Cardiac Monitoring Guideline

1. Purpose

The purpose of this document is to provide guidance for the initiation and removal of cardiac monitoring.

Due to the variety of clinical practice settings and patient cohorts in which cardiac monitoring is undertaken, this document does not stipulate absolute criteria but provides a point of reference for clinicians working in areas where cardiac monitoring is a component of patient assessment and in no way replaces the need for clinical assessment of the individual patient.

2. Applicability

This guideline applies to medical and nursing staff within the Armada Kalamunda Group caring for patients requiring cardiac monitoring.

3. Points of note

- The knowledge, skills and competencies of the clinical staff should be consistent with that required of those involved with the assessment, care and treatment of patients requiring cardiac monitoring. Nursing staff are to have relevant patient assessment, electrocardiogram (ECG) and cardiac rhythm interpretation skills.
- Clinical vigilance by the nursing and medical team to recognise life threatening, unstable arrhythmia or increase burden of symptomatic arrhythmia is essential for the appropriate medical management.
- Cardiac monitoring of the patient should be performed continuously. Patients can be monitored by the following means:
 - Direct visualisation – via bedside monitor
 - Indirect visualisation – via telemetry units with central monitoring station
- Staff caring for cardiac monitored patients should promptly respond to audible and visual alarms and review patient to determine if further escalation is required
- Patients to be prioritised for cardiac monitoring for the next 24 hours should be discussed and agreed during the afternoon handover
- Where demand exceeds capacity within the medical ward and/or Intensive Care Unit/High Dependency Unit (ICU/HDU), the decision regarding the need for cardiac monitoring and admission location should involve:
 - Discussion between MAU Registrar, ICU Registrar and ward coordinators
 - Escalation for Consultant-to-Consultant discussion if consensus is not reached, with input from Cardiologist as needed

- Escalation to DCS (in hours) or Executive on call (out of hours) if consensus regarding disposition cannot be reached by consultants
- A cardiac monitoring commencement instruction must be completed by medical team for all patients requiring cardiac monitoring into patient progress notes.
- To assist with patient access to cardiac monitored areas, there may be situations for consideration of nurse (NUM / ANUM) initiated removal of telemetry (non-acute) cardiac monitoring according to pre-determined criteria (for haemodynamically stable patients with no arrhythmias noted). This may be pre- indicated on the cardiac monitoring sticker.
- Patient notes needs to be updated by the medical staff upon cessation of cardiac monitoring
- The indication for cardiac monitoring must be re-assessed and documented by a senior treating Registrar/Consultant a minimum of twice daily.

4. Decision pathway

A decision pathway for the management of cardiac monitoring is included in Appendix 1.

For patients presenting to Armadale Health Service (AHS) with cardiac related symptoms, the Emergency Department (ED) senior clinician will decide if cardiac monitoring is required, with consultation with senior ICU clinician / General Medicine physician if required.

- If the patient requires urgent / immediate tertiary level care, the patient is to be transferred from AHS ED to the tertiary site via St John Ambulance.
- If the patient does not require tertiary level care, the cardiac monitoring is to be performed in the ICU/HDU or Medical Admissions Unit (MAU) (at AHS) with ongoing discussion with Royal Perth Hospital (RPH) medical teams as required.

Where patients are already admitted to AHS and experience a change in their condition that requires cardiac monitoring, the patient will either be

- Transferred to a tertiary site for tertiary level care, or
- Transferred to MAU/ICU/HDU at AHS as decided by the senior clinician, in consultation with the senior nurse

Any patient requiring transfer to a tertiary site will require a clinician-to-clinician discussion and handover.

5. Guidelines

These guidelines are based on a rating system derived from the American College of Cardiology Emergency Cardiac Care Committee and consist of 3 broad categories. The guidelines have been adapted for Armadale Hospital and can be tailored as per patient needs by the treating physicians.

- Class I - cardiac monitoring is strongly indicated.
- Class II - cardiac monitoring is advisable.
- Class III - cardiac monitoring is generally not indicated.

5.1 Class 1: Cardiac monitoring is strongly indicated

- Cardiac monitoring is not discontinued without a written medical order. The indication for ongoing cardiac monitoring must be reassessed and documented by a medical officer twice daily.
- An appropriately qualified/trained health professional, portable cardiac monitor and defibrillator to accompany the patient for all transfers.

	Monitoring Duration	Comment	Admission destination
Acute Coronary Syndromes (ACS): a) STEMI – follow pathway (Appendix 2) b) Very high risk			Transfer to RPH
c) High/intermediate risk ACS/ NSTEMI (Appendix 3)	48 hrs of cardiac monitoring, post presentation.	Discussion with cardiology	Admit to MAU
d) Multiple Risk factors, Troponin Negative	Consider cardiac monitoring up to 24hrs	Early risk stratification for future cardiac events with stress testing is advisable.	Admit to MAU
Life threatening arrhythmias a) Tachyarrhythmia - Post cardiac arrest – discuss need for transfer on a case by case basis. - VT/VF - AF or SVT with high risk of 1:1 conduction in presence of accessory pathway [1]. - Tachy-Brady syndrome *Arrhythmias leading to haemodynamic compromise	Continuous monitoring until reversible causes including metabolic derangements are rectified or transfer to tertiary centres for device implantation/EP intervention.	<i>*(Systolic BP below 90mmHg with symptoms of circulatory failure/cardiogenic shock)</i>	Transfer to RPH <i>If unable to transfer to RPH then admit to AHS ICU/HDU</i>
b) Symptomatic Brady arrhythmia - Sinus node dysfunction with Asystole of 3 seconds or evidence of chronotropic incompetence. 2nd or 3rd degree AV block - Syncope with evidence of conduction disease on ECG including trifascicular block,		As a bridging therapy Isoprenalline infusion with application of temporary/external pacer should be considered.	Transfer to RPH <i>If unable to transfer to RPH then admit to AHS ICU/HDU</i>

LBBB/RBBB with 1 st degree heart block.	Monitoring Duration	Comment	Admission destination
Congenital or Acquired Channelopathies • Prolonged QTc (>500msec) leading to NSVT or Torsades • CPVT • Intermediate to High risk long QT syndromes (LQTS) • Intermediate to High risk Brugada Syndrome		Consider early temporary pacing and/or isoprenaline infusion to keep the base HR above 80/min in case of acquired Torsades precipitated by prolonged QT secondary to bradycardia.	Transfer to RPH <i>If unable to transfer to RPH then admit to ICU/HDU</i>
Miscellaneous Indications a) During infusion and loading of • Anti-arrhythmics.e.g; amiodarone and flecainide • Fibrinolytics • Inotropes • Vasoactive Drugs • Pro-arrhythmic agents including phenytoin.	Continuous monitoring is indicated during the course of infusion and loading Prolonged monitoring with early cardiology referral is advised if there is any evidence of ongoing unstable arrhythmia		Admit ICU/HDU Fibrinolytics, Vasoactives, Overdose with arrhythmias
b) Overdose of pro-arrhythmics • Leading to ECG changes with or without life threatening arrhythmias	Duration of monitoring based on risk stratification i.e. baseline LV function, type of drug, time since ingestion and total dose		Admit MAU Amiodarone, Overdose without arrhythmias
c) Moderate to Large Pericardial effusion d) Infective Endocarditis with moderate to severe valve dysfunction or valve abscess e) Cardiac contusion			Admit RPH
Acute and severe electrolytes disturbances (post initial treatment)	Monitor until the acute electrolyte disturbance is corrected and	Risk stratification of patients for recurrence of arrhythmia including	Admit ICU/HDU K >6 with ECG changes

	Monitoring Duration	Comment	Admission destination
K $\geq$ 6.0 with ECG changes Mg $\leq$ 0.4 Corrected Ca. $<$1.9 Severe Hyponatremia (Na $\leq$120mmol/L)	there are no related arrhythmias present	LV function assessment by echo is advised.	Na$<$120 Mg $<$ 0.4 with ECG changes Ca $<$ 1.7 Otherwise admit to MAU

5.2 Class II: Cardiac monitoring is advisable as per risk benefit ratio

- Shift Coordinator may discontinue cardiac monitoring at the completion of the recommended monitoring period if the patient's clinical condition is stable and there is no written order from a medical officer to continue cardiac monitoring beyond this period.

Indications for monitoring	Duration	Comments	Admission destination
Acute Coronary Syndrome: Low risk Trop negative ACS	Monitor until second troponin is available If 2nd Tn is positive manage as Class I If 2nd Tn is negative, consider early stress test	Patients for comfort treatment can be excluded from monitoring	D/C home <i>if patient already admitted treat on ward with consult to physician if required</i>
Arrhythmias a) Non-life threatening Tachyarrhythmia New onset SVT or AF with fast V rate (resting$>$140/min) and evidence of systemic compromise – hypotension, oliguria, or desaturation Chronic AF with faster than usual V rate b) Asymptomatic Bradycardia with ECG evidence of conduction disease			Admit MAU / ICU/HDU Discussion between ICU and Gen Med re monitoring requirements
Acute stroke (ambulance bypass in place)	12-24 hrs monitoring		Transfer to RPH

Indications for monitoring	Duration	Comments	Admission destination
Syncope of unknown cause with normal ECG	24-48 hrs monitoring		Admit MAU

5.3 Class III

- Terminally ill patients with Goals of Patient Care Patients for comfort treatment deemed unsuitable for any intervention
- The issue of management of patients with Goals of Patient Care Active ward-based treatment is the subject of ongoing discussion between ICU and General Medicine and agreement
- Chronic stable atrial fibrillation

6. Escalation process

Where there is a difference of opinion regarding a decision for the right patient in the right place at the right time, this is to be escalated:

In hours to the referring Consultant /NUM/ANUM, with further escalation to the AKG DCS if required.

After hours Consultant on call / Nurse Manager, with further escalation to the AKG Executive on call where required.

7. Supporting information

The following documents support the implementation of this policy:

- AKG [After Hours Communication Huddle Procedure](#)
- AKG [Senior Response On-Call Support Guideline](#)

8. Compliance, monitoring and evaluation

Monitoring of the adherence to the requirements of this guideline will be facilitated via a quarterly audit of a minimum of 30 medical records of patients that are indicated for cardiac monitoring within the Medical Admissions Unit. The audit is at a minimum to report on the following indicators:

- Usage rates of commencement sticker
- Usage rate of cessation sticker
- Proportion of stickers completed in accordance with the requirements of this guideline
- Adherence to assessment, monitoring and de-monitoring requirements for patients requiring cardiac monitoring.

9. Relevant standards

National Safety and Quality Health Service (NSQHS) Standards (second edition):

- Standard 8 – Action 8.4
- Standard 8 – Action 8.13
- Standard 6 – Action 6.4

10. References

1. Victorian Cardiac Clinical Network: Cardiac monitoring /telemetry of adult patients in Victorian Public hospitals Draft 5 2013.
2. New South Wales (NSW) Health 2008, Policy directive: Cardiac monitoring in adult cardiac patients in public hospitals in NSW.
3. Therapeutic guidelines Ltd. 2015
4. Page, R.L., et al., 2015 ACC/AHA/HRS Guideline for the Management of Adult Patients With Supraventricular Tachycardia. A Report of the American College of Cardiology/American Heart Association Task Force on Clinical Practice Guidelines and the Heart Rhythm Society, 2016. 133(14): p. e506-e574.

11. Document owner

Enquiries relating to this guideline may be directed to:

Title: Head of Department
Department: General Medicine

12. Review

This guideline will be reviewed and evaluated as required to ensure relevance and currency. At a minimum it will be reviewed within one (1) year after first issue and at least every three (3) years thereafter.

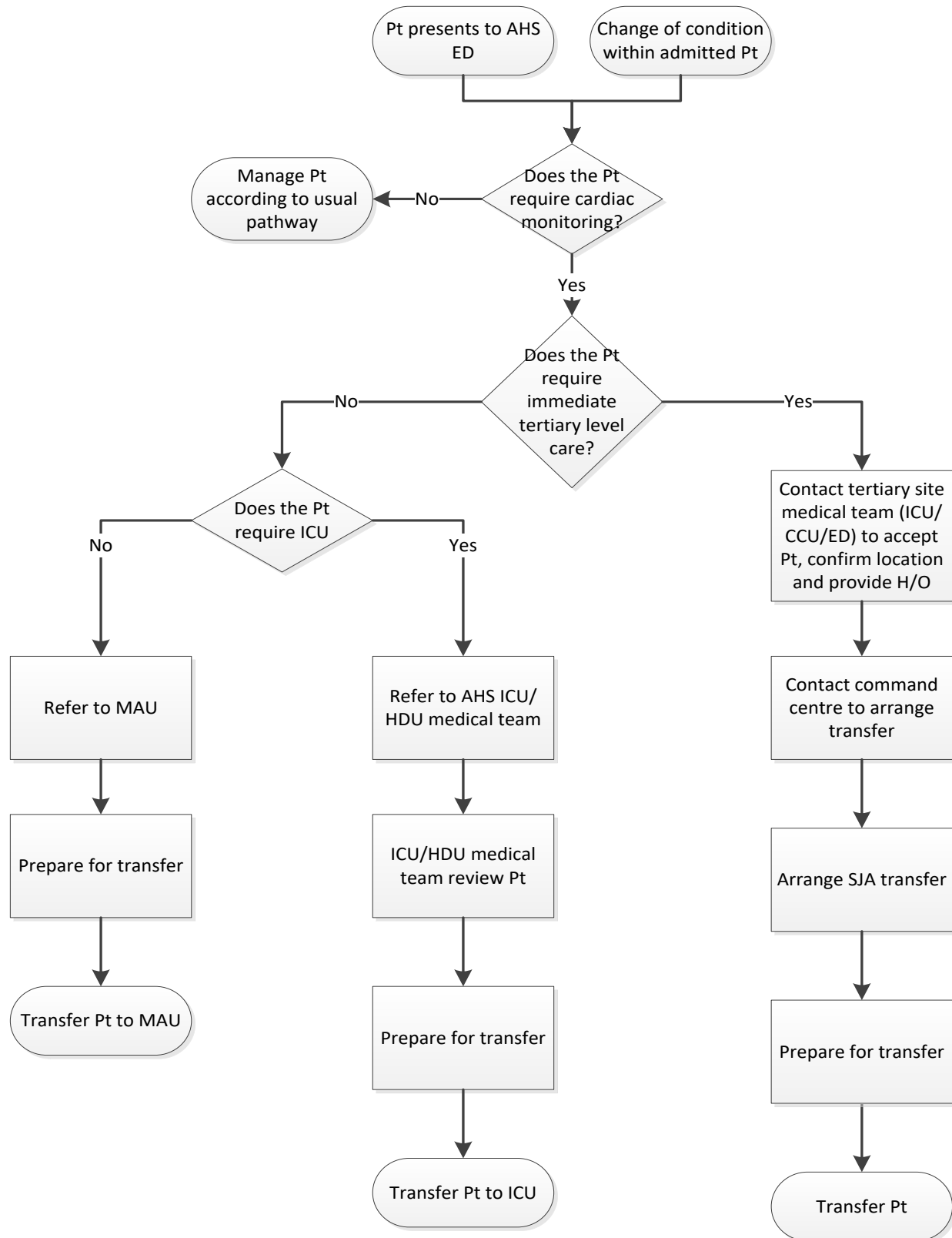
Version	Effective from	Effective to	Amendment(s)
V3	06/11/2023	06/11/2026	Scheduled review

13. Authorisation

This guideline has been authorised and issued by Director Clinical Services

Authorised by	Dr Alison Parr – Director Clinical Services
Authorisation date	06/11/2023
Published date	06/11/2023

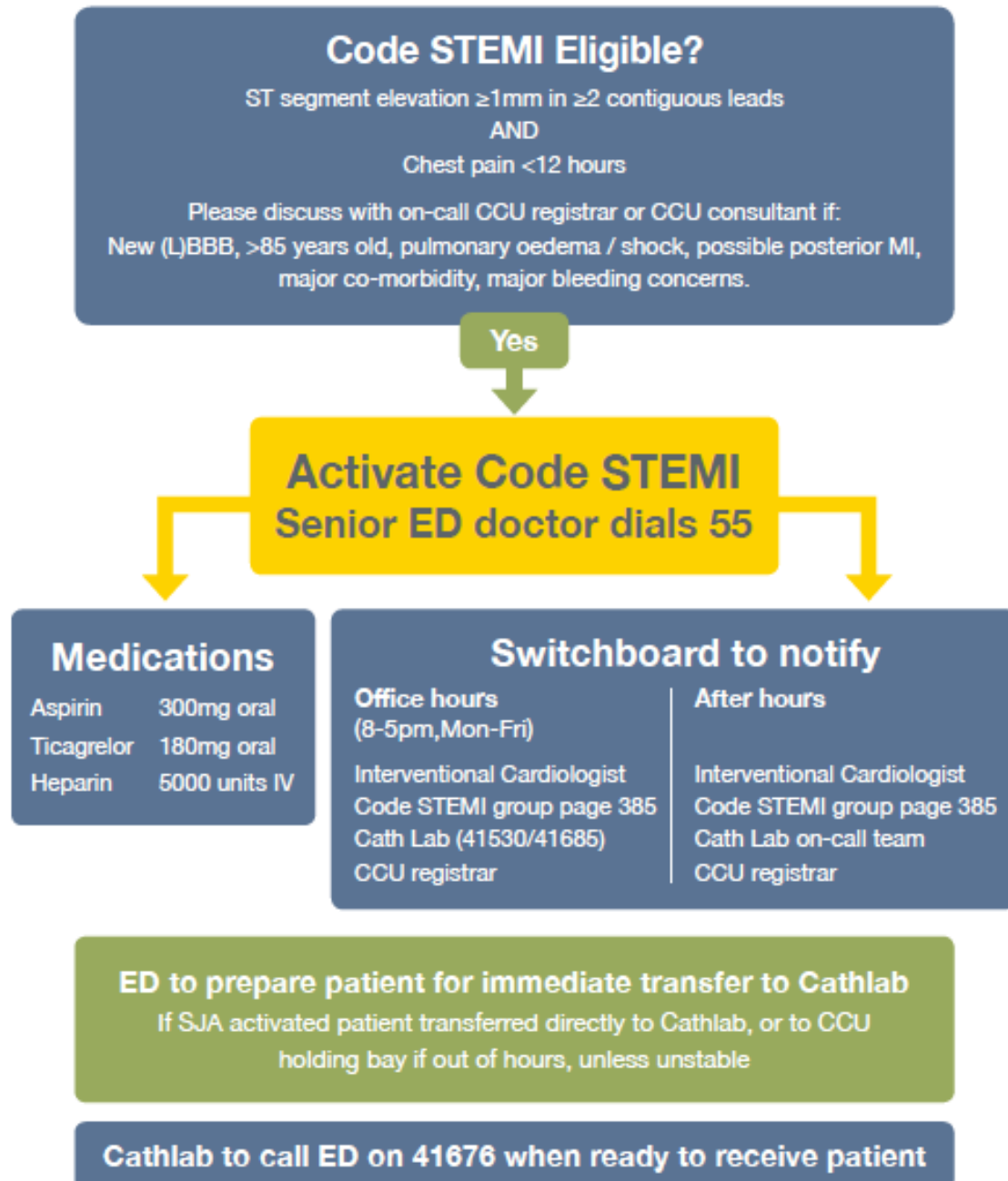
Appendix 1: Decision pathway for management of patients requiring cardiac monitoring



Appendix 2: STEMI activation pathway

RPH / SJA STEMI Activation Pathway

12-lead ECG demonstrates STEMI (confirmed by senior ED doctor)



Appendix 3: Suspected ACS pathway

Suspected ACS Pathway

